

1 KPI Monitor

1.1 General

App name	KPI monitor
Short name of app	KPI
Function authorization	sma.kpim

When you open the application *KPI monitor*, you get a clear an overview of the KPIs defined for the factory/site. You can also get more detailed information and drill down the different KPIs in relation to the integrated dimensions and the relevant object "machine".

The *KPI monitor* currently provides the following KPIs:

- OEE and its components

The *KPI monitor* (start screen) shows the OEE and its components for all machines (site) as a bar chart in the *OEE* section.

- OEE
- OEE performance
- OEE availability
- OEE quality

The calculation is based on the formulas defined in HYDRA.

If you click the OEE section, you are forwarded to the detail application OEE.

- Rate of capacity utilization (utilization efficiency)

The KPI monitor (start screen) shows the current KPI rate of capacity utilization for all machines (site) in a horizontal bar chart in the *Rate of capacity utilization* section.

If you click the *Rate of capacity utilization* section, you are forwarded to the detail application *Rate of capacity utilization*.

- Produced quantities

The KPI monitor (start screen) shows the produced yield and scrap quantity for all machines (site) in a horizontal bar chart in the *Produced quantities* section.

If you click the *produced quantities* section, you are forwarded to the detail application *produced quantities*.

- Scrap rate

The KPI monitor (start screen) shows the scrap rate for all machines (site) in a bar chart in the *scrap rate* section.

If you click the *scrap rate* section, you are forwarded to the detail application *scrap rate*.

- Production times and downtimes

The KPI monitor (start screen) shows the production times and downtimes for all machines (site) in a pie chart in the *production and downtimes* section.

If you click the *production and downtimes* section, you are forwarded to the detail application *production times and downtimes*.

- Failure report

The KPI monitor (start screen) shows the produced yield and failure (defective) quantity for all machines (site) in a horizontal bar chart in the *failure report* section. The failure quantity results from the number of recorded failure types and their weighting.

If you click the *failure report* section, you are forwarded to the detail application.

1.2 Detail application OEE

Graphic presentation (diagram) of the KPI OEE and its components for the object selected in the list.

Please note: The list only shows the workplaces for which data is available to calculate the OEE.

Functions of the detail application

- Changing the display layout

You can switch between the following display options:

- Horizontal bar chart
- Vertical bar chart

- Changing the dimension displayed

You can display KPIs for the following dimensions in the detail view:

- Workplace group/machine group
- Cost center
- Workplace/machine

1.3 Detail application Rate of capacity utilization

Graphic presentation (diagram) of the KPI *rate of capacity utilization* (green) for the object selected in the list.

The KPI *rate of capacity utilization* is calculated as follows:

$$\text{Rate of capacity utilization} = \frac{RPA_{11}}{\sum RPA_{1-11}}$$

Please note: The list only shows the workplaces for which data is available to calculate the *rate of capacity utilization*.

Functions of the detail application

- Changing the display layout

You can switch between the following display options:

- Horizontal bar chart
- Vertical bar chart
- Pie chart

- Changing the dimension displayed

You can display KPIs for the following dimensions in the detail view:

- Workplace group/machine group
- Cost center
- Workplace/machine

1.4 Detail application *Produced quantities*

Graphic presentation of the produced yield and scrap quantities each in the "primary" quantity unit.

Please note: The list only shows the workplaces for which data is available to calculate the produced quantities.

Functions of the detail application

- Changing the display layout

You can switch between the following display options:

- Horizontal bar chart
- Vertical bar chart

- Changing the dimension displayed

You can display KPIs for the following dimensions in the detail view:

- Workplace group/machine group
- Cost center
- Workplace/machine

1.5 Detail application *Scrap rate*

Graphic presentation (diagram) of the KPI *scrap rate* for the object selected in the list. The KPI *scrap rate* is calculated as follows:

$$\text{Scrap rate} = \frac{\text{Scrap}_{\text{Primary}}}{\text{Scrap}_{\text{Primary}} + \text{Yield}_{\text{Primary}}}$$

Please note: The list only shows the workplaces for which data is available to calculate the scrap rate.

Functions of the detail application

- Changing the display layout

You can switch between the following display options:

- Horizontal bar chart
- Vertical bar chart
- Pie chart

- Changing the dimension displayed

You can display KPIs for the following dimensions in the detail view:

- Workplace group/machine group
- Cost center
- Workplace/machine

1.6 Detail application *Production times and downtimes*

Graphic presentation (diagram) of production times and downtimes for the object selected in the list. Production time refers to the time posted to the resource performance account (RPA) 11 and downtime refers to the durations posted to the resource performance accounts (RPA) 1-10.

Please note: The list only shows the workplaces for which data is available to calculate production times and downtimes.

Functions of the detail application

- Changing the display layout

You can switch between the following display options:

- Horizontal bar chart
- Vertical bar chart

- Changing the dimension displayed

You can display KPIs for the following dimensions in the detail view:

- Workplace group/machine group
- Cost center
- Workplace/machine

1.7 Detail application *Failure report*

Graphic presentation of the produced yield in the "primary" quantity unit and the failure (defective) quantity of quality inspections. The failure quantity results from the number of recorded failure types and their weighting.

Please note: The list only shows the workplaces for which data is available to calculate the produced quantities.

Functions of the detail application

- Changing the display layout

You can switch between the following display options:

- Horizontal bar chart
- Vertical bar chart

- Changing the dimension displayed

You can display the KPI *failure rate in ppm* for the following dimensions in the detail view:

- Workplace group/machine group
- Cost center
- Workplace/machine

The KPI *failure rate in ppm* is calculated as follows:

$$\text{Scrap rate} = \frac{\text{Failure quantity}}{\text{Scrap}_{\text{Primary}} + \text{Yield}_{\text{Primary}}} * 1.000.000$$

1.8 General – KPI Monitor

Selection/filter options of the *KPI monitor*

▼ The *KPI monitor* provides the following selection options:

Date from / to

The selection criterion *Date from/to* specifies the period of time you want to evaluate.

Filter mode time/shift

Use the *filter mode* to specify whether the selection is based on a time or shift specified.

Shift

If you use the selection criterion *Shift*, you select shifts to specify the evaluated period of time (date from ... to ...). If the selection period exceeds the period for the online data area, the system automatically selects the medium-term data area.

Time from/to

If you use the selection criterion *Time*, you select a time (from... to...) to specify the evaluated period of time (date from ... to ...). If the selection period exceeds the period for the online data area, the system automatically selects the medium-term data area.



When you open the KPI monitor, the system uses the default selection for the display, which is defined in the application settings. If you change the selection, this selection applies for the

duration of the session.

Use the button *Reset* to reset the selection to the default selection defined in the application settings.

If you click a detail application, the selection criteria are used for the detail application.



The *Failure report* cannot show any evaluations based on shifts. If you have made a selection by shift, this report does not display any key figure then. The KPIs in the *Failure report* are only available if you have made a selection by *Time from/to*.

Application settings in the KPI monitor

The *KPI monitor* provides the following application settings. These settings are used to predefine the selection criteria used to open the *KPI monitor*:

Default date in filter

Use the field *Default date in filter* to assign the current date (date when KPI monitor is called) to the selection field *Date from/to* or a calculated date with a relative specification. If you specify a relative date, the following options are available:

- Number with the signs + and -
- Time period with the following options: days/weeks/months/years

Example:

Date of call	Default date in filter (relative)	Transfer to the selection field <i>Date from/to</i>
15.05.2019	from: -2 days to: + 3 days	from: 13.05.2019 to: 18.05.2019

Preset filter mode

If you use the field *Preset filter mode*, the selection field *filter mode* is predefined, which specifies whether the selection is based on a time or shift specified.

Default shift

Use the field *Default shift* to predefine the selection field *Shift*, which specifies the evaluated period of time (date from ... to ...) using the shifts selected.

Time from/to

Use the field *Time from/to* to predefine the selection field *Time from/to*, which specifies the evaluated period of time (date from ... to ...) using the time (from... to...) selected.

1.9 General – Detail applications

Selection/filter options in the detail applications

▼ The detail applications provide the following selection options:

Resource

Enter the resource number (workplace number) to select from the workplaces.

Group

Workplace group

Cost center

Cost center of workplaces

Date from / to

The selection criterion *Date from/to* specifies the period of time you want to evaluate.

Filter mode time/shift

Use the *filter mode* to specify whether the selection is based on a time or shift specified.

Shift

If you use the selection criterion *Shift*, you select shifts to specify the evaluated period of time (date from ... to ...). If the selection period exceeds the period for the online data area, the system automatically selects the medium-term data area.

Time from/to

If you use the selection criterion *Time*, you select a time (from... to...) to specify the evaluated period of time (date from ... to ...). If the selection period exceeds the period for the online data area, the system automatically selects the medium-term data area.

If you change the selection, this selection applies for the duration of the session and is also used when you call the KPI monitor the next time.



The detail application *Failure report* cannot show any evaluations based on shifts. The selection fields *Filter mode time/shift* and *Shift* are therefore not available.

Field descriptions – List

KPI

Numeric presentation of the KPI

Machine

Machine number and short description of the machine

Full name of machine

Full name of the machine

Field descriptions – Detail**Machine**

Machine number and short description of the machine

Graphic

Graphic display of the relevant KPI in a diagram